

IF440: Artificial Intelligence

Topic 07: Hidden-Markov Model

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Course Sub Learning Outcomes (Sub-CLO):



SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).

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Review

- 1 Markov Chain
- 2 Transition Matrix
- 3 Example of Applied Markov Chains
- 4 Practices



Outline

1 What is an HMM?

2 HMM Formalism

3 Example

4 Viterbi Algorithm



Pre-Class Assignment



- Stanford University <https://www.youtube.com/watch?v=mNSQ-prhgsW>

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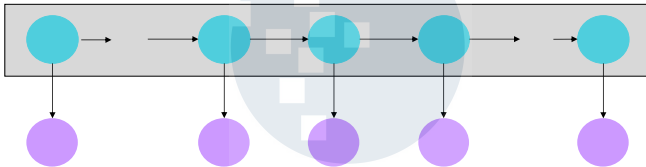


What is an HMM?

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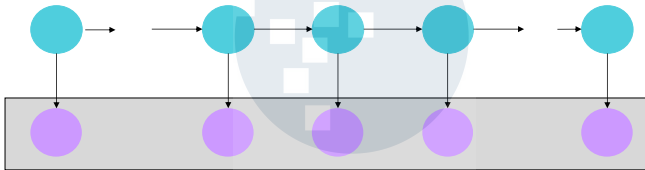
- Graphical Model
- Circles indicate states
- Arrows indicate probabilistic dependencies between states

WHAT IS AN HMM?



- Blue nodes are **hidden states**
- Dependent only on the previous state
- “The past is independent of the future given the present.”

WHAT IS AN HMM?



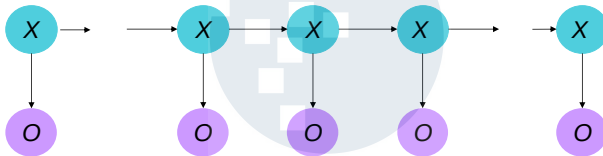
- Purple nodes are **observed states**
- Dependent only on their corresponding hidden state



HMM Formalism

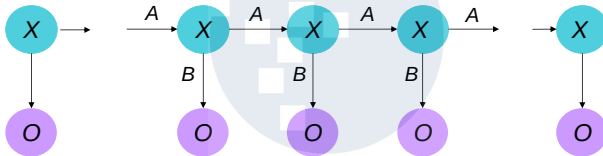
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HMM FORMALISM



- $\{X, O, \Pi, A, B\}$
- $X : \{x_1 \dots x_N\}$ are the values for the hidden states
- $O : \{o_1 \dots o_M\}$ are the values for the observed states

HMM FORMALISM



- $\{X, O, \pi, A, B\}$
- $\pi = \{\pi_i\}$ are the initial state probabilities
- $A = \{a_{ij}\}$ are the state transition probabilities
- $B = \{b_{ik}\}$ are the observation state probabilities



Example

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EXAMPLE



Alice

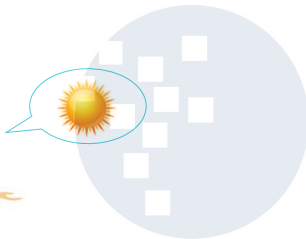


Bob

EXAMPLE



Alice



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Bob



EXAMPLE



Alice



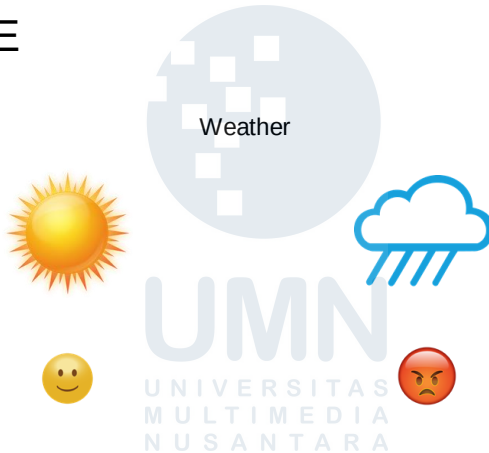
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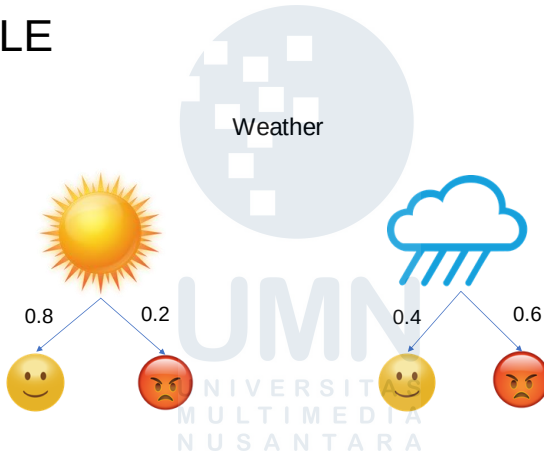
Bob



EXAMPLE



EXAMPLE



EXAMPLE



Alice



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Bob



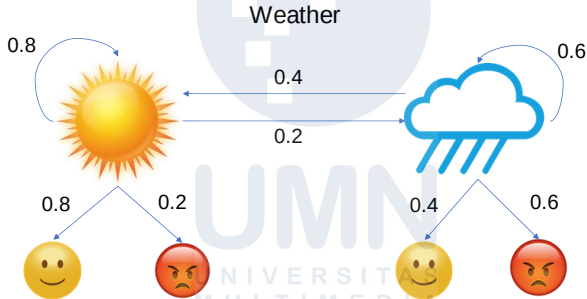
EXAMPLE

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
						

Alice inferences

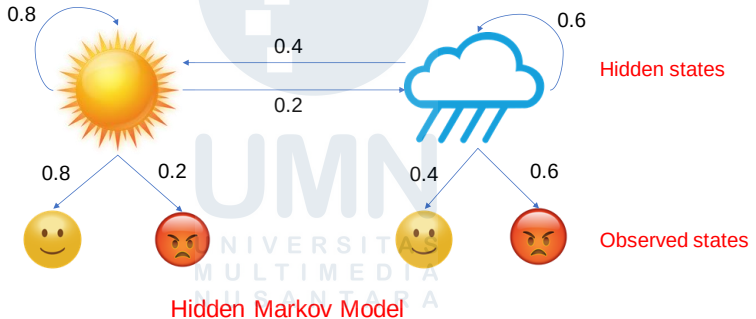


EXAMPLE

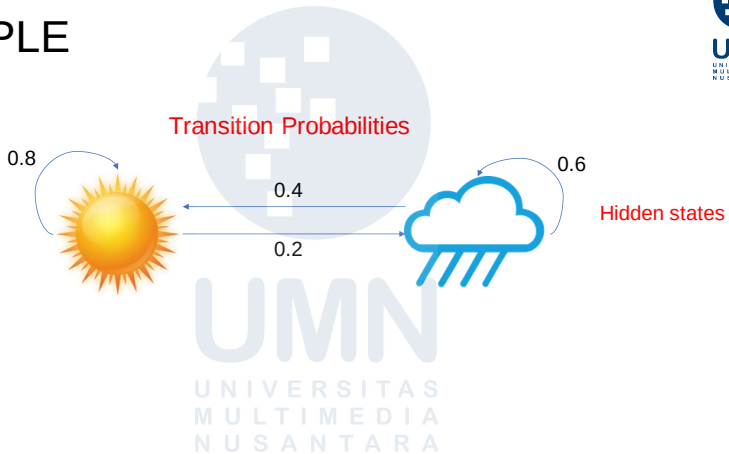


Hidden Markov Model

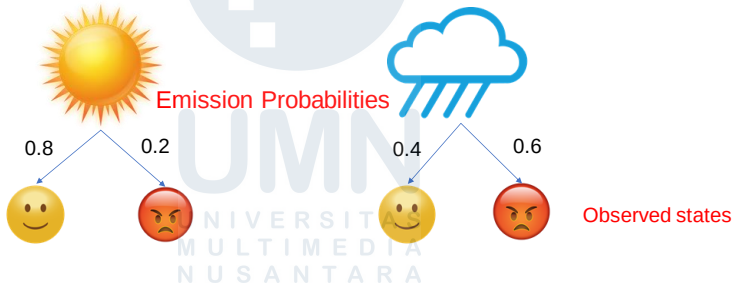
EXAMPLE



EXAMPLE



EXAMPLE

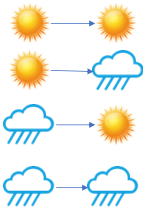


EXAMPLE: Questions

1. **How did we find these probabilities?**
2. What's the probabilities that a random day is Sunny or Rainy?
3. If Bob is happy today, what's the probability that it's Sunny or Rainy?
4. If for three days Bob is happy, grumpy, happy, what was the weather?

EXAMPLE: Questions

1. How did we find these probabilities? **Observed data from the past**



8 0.8

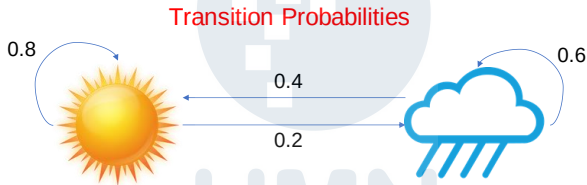
2 0.2

2 0.4

3 0.6

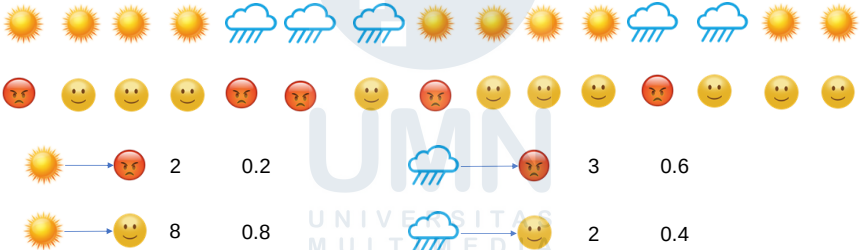
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EXAMPLE: Questions

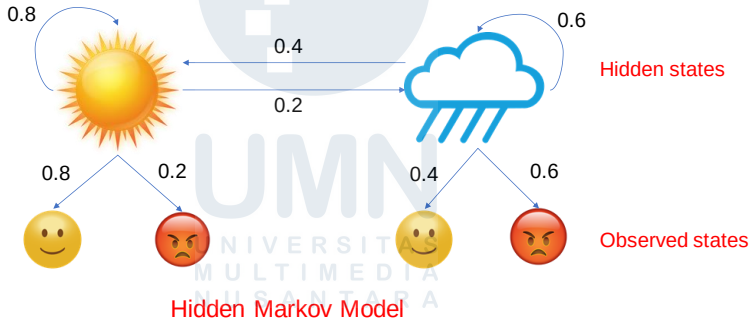


EXAMPLE: Questions

1. How did we find these probabilities? **Observed data from the past**



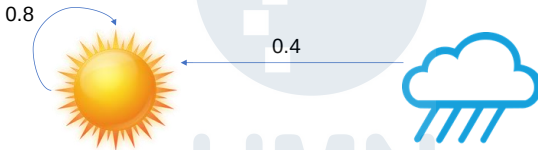
EXAMPLE: Questions



EXAMPLE: Questions

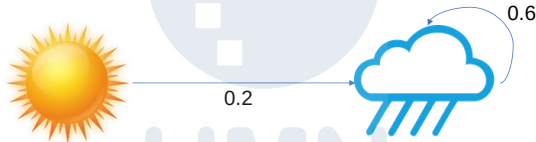
- ~~1. How did we find these probabilities?~~
- 2. What's the probabilities that a random day is Sunny or Rainy?**
3. If Bob is happy today, what's the probability that it's Sunny or Rainy?
4. If for three days Bob is happy, grumpy, happy, what was the weather?

EXAMPLE: Questions



$$\begin{aligned}
 P(\text{Sunny}) &= P(\text{Sunny}|\text{Sunny}) P(\text{Sunny}) + P(\text{Sunny}|\text{Rainy}) P(\text{Rainy}) \\
 &= 0.8 P(\text{Sunny}) + 0.4 P(\text{Rainy})
 \end{aligned}$$

EXAMPLE: Questions



$$P(\text{Rainy}) = P(\text{Rainy}|\text{Rainy}) P(\text{Rainy}) + P(\text{Rainy}|\text{Sunny}) P(\text{Sunny})$$

$$= 0.6 P(\text{Rainy}) + 0.2 P(\text{Sunny})$$

$$P(\text{Rainy}) + P(\text{Sunny}) = 1$$

EXAMPLE: Questions



$$P(\text{Sunny}) = 2/3$$



$$P(\text{Rainy}) = 1/3$$

$$P(\text{Sunny}) = 0.8 P(\text{Sunny}) + 0.4 P(\text{Rainy})$$

$$0.2 P(\text{Sunny}) = 0.4 P(\text{Rainy})$$

$$1 P(\text{Sunny}) = 2 P(\text{Rainy})$$

EXAMPLE: Questions

- ~~1. How did we find these probabilities?~~
- ~~2. What's the probabilities that a random day is Sunny or Rainy?~~
- 3. If Bob is happy today, what's the probability that it's Sunny or Rainy?**
4. If for three days Bob is happy, grumpy, happy, what was the weather?

EXAMPLE: Questions

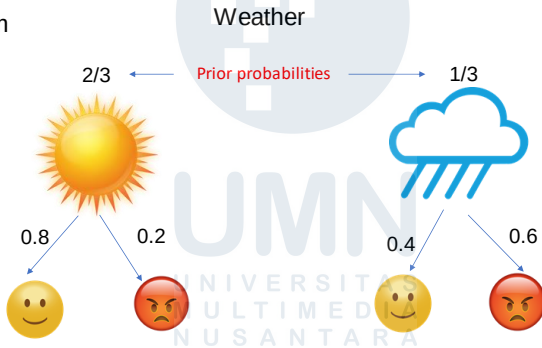
Bayes Theorem

Today, Feeling Happy



EXAMPLE: Questions

Bayes Theorem



EXAMPLE: Questions

Bayes Theorem

$$P(S|H) = \frac{P(H|S) \cdot P(S)}{P(H)}$$

$$P(H) = P(H|S) \cdot P(S) + P(H|R) \cdot P(R)$$

$$P(R|H) = \frac{P(H|R) \cdot P(R)}{P(H)}$$

EXAMPLE: Questions

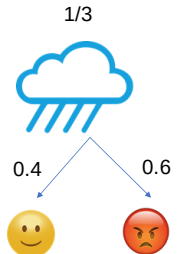
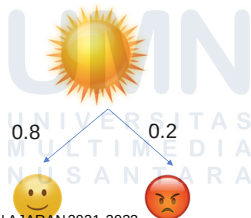
Bayes Theorem

$$P(H) = P(H|S) \cdot P(S) + P(H|R) \cdot P(R)$$

$$= 0.8 \cdot \frac{2}{3} + 0.4 \cdot \frac{1}{3}$$

$$= 0.536 + 0.132$$

$$= 0.668$$



EXAMPLE: Questions

Bayes Theorem

$$P(S|H) = \frac{P(H|S) \cdot P(S)}{P(H)}$$

$$= \frac{0.8 \cdot 2/3}{0.668}$$

$$= 0.8$$

$$P(R|H) = \frac{P(H|R) \cdot P(R)}{P(H)}$$

$$= \frac{0.4 \cdot 1/3}{0.668}$$

$$= 0.2$$

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EXAMPLE: Questions

- ~~1. How did we find these probabilities?~~
- ~~2. What's the probabilities that a random day is Sunny or Rainy?~~
- ~~3. If Bob is happy today, what's the probability that it's Sunny or Rainy?~~
- 4. If for three days Bob is happy, grumpy, happy, what was the weather?**

EXAMPLE:
If happy-grumpy, what's the weather?

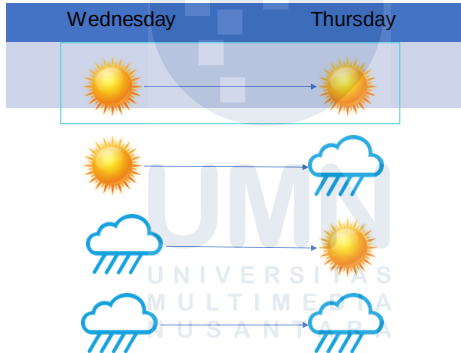
Wednesday	Thursday
	



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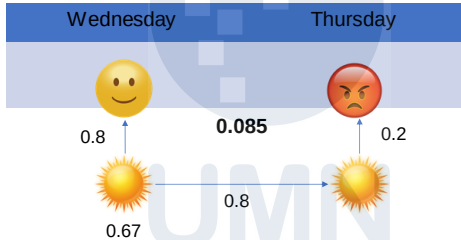
EXAMPLE:

If happy-grumpy, what's the weather?



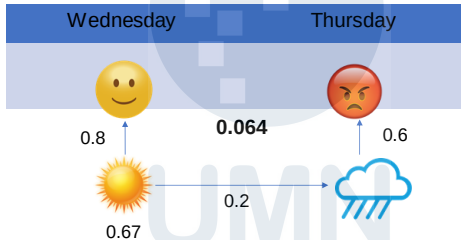
EXAMPLE:

If happy-grumpy, what's the weather?



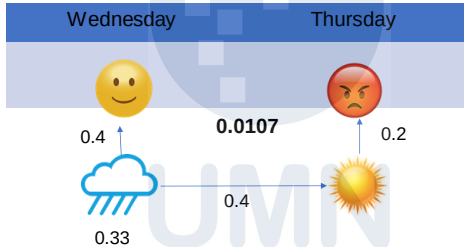
EXAMPLE:

If happy-grumpy, what's the weather?



EXAMPLE:

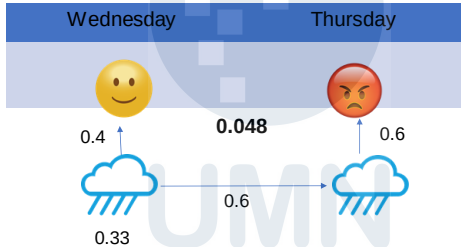
If happy-grumpy, what's the weather?



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EXAMPLE:

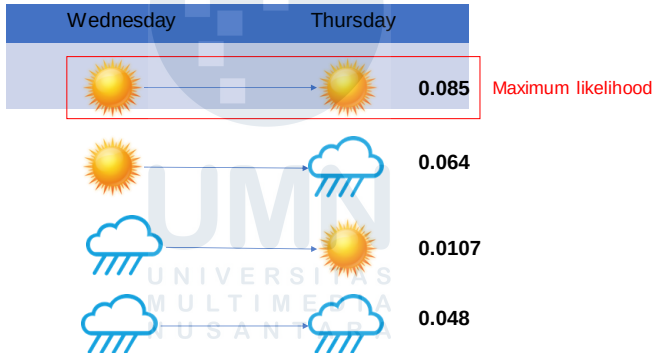
If happy-grumpy, what's the weather?



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EXAMPLE:

If happy-grumpy, what's the weather?



EXAMPLE:

If happy-grumpy-happy, what's the weather?

Wednesday	Thursday	Friday
		



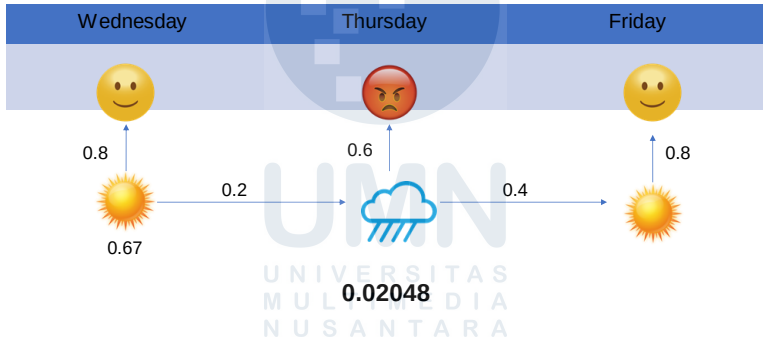
EXAMPLE:

If happy-grumpy-happy, what's the weather?



EXAMPLE:

If happy-grumpy-happy, what's the weather?



EXAMPLE: If 6 days?

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					

Viterbi Algorithm

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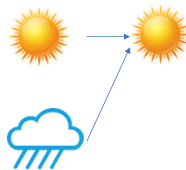
Viterbi Algorithm

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VITERBI ALGORITHM

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					

Best path up to here



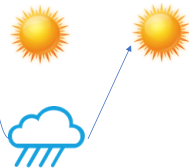
VITERBI ALGORITHM

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					

Best path up to here

Compare with

Best path up to here



VITERBI ALGORITHM

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					

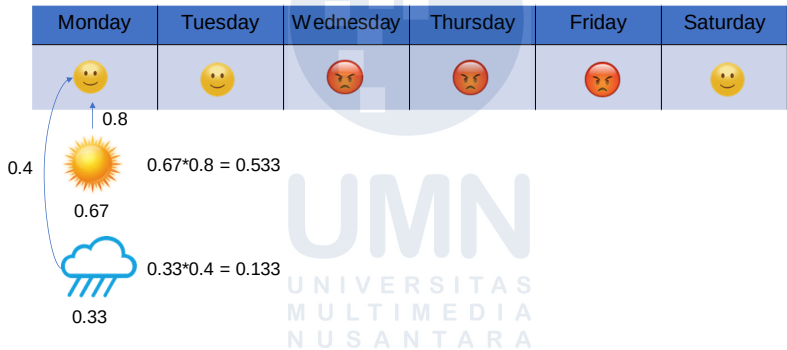
Best path up to here

Compare with

Best path up to here

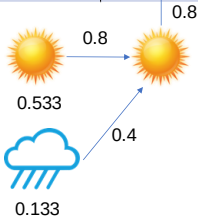


VITERBI ALGORITHM



VITERBI ALGORITHM

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					



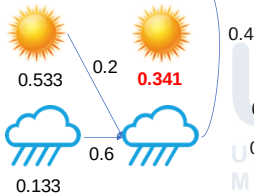
$$0.533 \times 0.8 \times 0.8 = \mathbf{0.341}$$

$$0.133 \times 0.4 \times 0.8 = 0.0425$$

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VITERBI ALGORITHM

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					

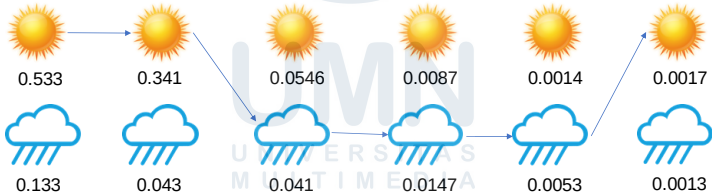


$$0.533 \times 0.2 \times 0.4 = 0.043$$

$$0.133 \times 0.6 \times 0.4 = 0.032$$

VITERBI ALGORITHM

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					



VITERBI ALGORITHM

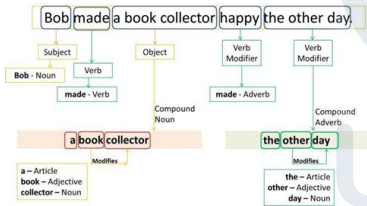
Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					



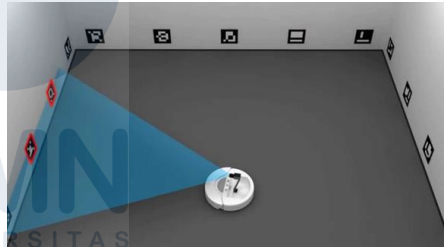
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APPLICATIONS

Part of speech tagging



Robot localization



APPLICATIONS

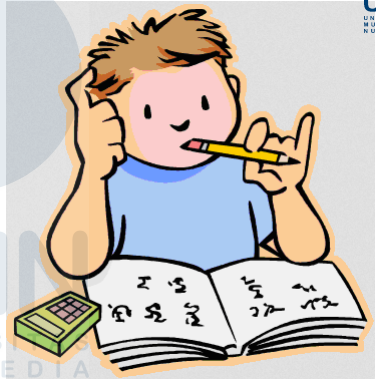
Speech recognition



Genetics

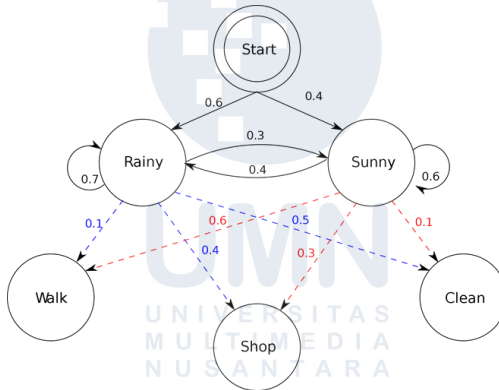


PRACTICE



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PRACTICE



PRACTICE

- 1 If Bob is **walking** today, what's the probability that it's Sunny or Rainy?
- 2 If Bob is **shopping** today, what's the probability that it's Sunny or Rainy?
- 3 If Bob is **cleaning** today, what's the probability that it's Sunny or Rainy?
- 4 If for two days Bob is **walking, shopping**, what was the weather?
- 5 If for FIVE days Bob is **walking, shopping, walking, shopping, cleaning**, what was the weather? Find with Viterbi algorithm.

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Example



<https://www.mygreatlearning.com/blog/pos-tagging/>

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References

- [1] S. J. Russell, P. Norvig, and E. Davis, **Artificial intelligence: a modern approach**, 3rd ed. Prentice Hall, 2010.



Next Week

■ Midterm Exam



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Vision and Mission

Vision

Becoming a leading Informatics Undergraduate Study Program that produces graduates with international insight who are competent in the field of computer science, have an entrepreneurial spirit, and have noble character.



Mission

- 1 Organising learning with the best technology and curriculum and supported by professional teaching staff.
- 2 Carry out research activities in the field of informatics to advance information science and technology.
- 3 Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.

